RescueSwarm: An Indigenously Sourced Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Research and Development Funding Proposal

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***Abstract*—India recorded 5.4 million internal displacements from disasters in 2024, the highest figure in twelve years, and the binding constraint on survivor outcomes is the speed with which casualties are located and reached. Ground and boat-based search is slow and hazardous; helicopter search is fast but scarce and expensive. This proposal describes RescueSwarm, a three-aircraft autonomous unmanned aerial system that surveys a designated area, detects and geolocates survivors, and delivers a standardised relief kit to each, under a single operator and without any reliance on cellular or internet infrastructure. The system is designed around a deliberate constraint: maximal indigenous sourcing, both because supply-chain sovereignty is a national priority and because domestic lead times are roughly half those of imports, which directly buys flight-test days inside a schedule-limited programme. I present the system architecture, the sizing and error models that fix its design point, the simulation evidence obtained to date, a market-verified cost basis, and a phased work plan. I am explicit throughout about which figures are measured, which are modelled, and which remain assumed; no aircraft has yet flown, and the proposal is structured so that a reviewer can distinguish demonstrated capability from projected capability without reading the source material. The requested funding is INR 28.74 L over an eight-month programme.**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, coverage path planning, RTK geolocation, edge inference, indigenous manufacturing, disaster response**

I. Introduction and Problem Statement

A. The operational gap

India records among the highest disaster-displacement figures in the world. The Internal Displacement Monitoring Centre reports **5.4 million internal displacements in India during 2024** — the highest in twelve years — of which 2.4 million were triggered by a single monsoon flood season in Assam [8]. The scale of the response problem is therefore not in question; what is in question is how quickly the people displaced by it can be found.

Post-flood search and rescue in India is dominated by a single quantity: the elapsed time between the onset of the event and the moment a survivor is both *located* and *reached*. Nationally deployed capability addresses the second of these far better than the first. Boat teams and ground parties can reach a survivor once told where to go, but they search slowly, at high risk to the crew, and with poor coverage of debris fields and rooftops. Rotary-wing aircraft search quickly but are scarce, costly per flight hour, and are typically committed to evacuation rather than to systematic area survey.

The consequence is that *localisation* is the bottleneck. A search asset that can systematically cover an assigned polygon, detect human figures in cluttered post-flood imagery, and report each detection as a coordinate accurate to a few metres would materially change the tasking of every downstream resource.

B. Why a swarm rather than a single aircraft

A single multirotor with a useful payload and a fifteen-minute endurance covers a disappointing area. Coverage scales with the number of simultaneously operating aircraft, but so does operator workload, and a system requiring one pilot per aircraft does not deploy well in a disaster. The engineering problem is therefore not simply flight, but *coordinated autonomous flight under a single operator*: area decomposition, task allocation, deconfliction, consolidated reporting, and graceful degradation when the command link is lost.

C. Why indigenous sourcing is a design constraint, not a slogan

I treat domestic sourcing as a first-class design constraint for two reasons, only one of which is political.

The first is sovereignty: a disaster-response asset whose critical components have a single foreign supplier is a strategic liability, and national policy correctly prioritises domestic capability.

The second is schedule, and in a programme of this length it dominates. My supplier survey puts typical import lead times at four to six weeks against two to three weeks domestically. In a programme whose binding constraint is an eight-week flight-test window, that difference is not a procurement detail — it is the difference between two test campaigns and one. Indigenisation, in this programme, buys flying days.

D. Contributions of the proposed work

1) A three-aircraft autonomous survey-and-delivery system operable by a single operator with no network dependency, sized and costed against market-verified Indian components.
2) A geolocation error budget that treats survivor coordinate accuracy as the primary system output, and an architecture that meets it using a team-owned RTK base rather than any external correction service.
3) A coordination scheme — static equal-area decomposition with boustrophedon coverage and greedy claim-and-lock task allocation — selected for determinism and verifiability over optimality.
4) An evidence discipline in which every published number regenerates from its model in continuous integration, and in which measured, modelled and assumed quantities are separated by construction.

II. Objectives and Scope

A. Primary objectives

O1 Survey a designated area autonomously with three cooperating aircraft under one operator and one ground control station.

O2 Detect human figures in nadir imagery at a ground sample distance sufficient for reliable classification, with onboard inference and multi-frame temporal fusion.

O3 Report each detection as a geodetic coordinate with a circular error probable within a few metres.

O4 Deliver a standardised relief kit to each confirmed survivor location by controlled free-fall release.

O5 Operate with no cellular, internet or cloud dependency at any point in the mission.

O6 Maximise value-weighted indigenous content across the priced bill of materials without compromising O1–O5.

B. Explicit non-objectives

I state these because proposals of this kind are frequently over-scoped, and because each exclusion is a deliberate engineering decision rather than an omission.

- **Not a casualty-evacuation platform.** The payload class is a relief kit, not a person, and no growth path to the latter is claimed.
- **Not beyond-visual-line-of-sight certified.** The design operates within a bounded geofence; regulatory expansion is future work.
- **Not a night or all-weather system.** The perception stack is visible-band. Thermal imaging is identified as the highest-value future extension but is outside this programme.
- **Not a survivor-triage system.** The system reports position and delivers a kit; it makes no medical assessment.

III. Related Work and State of the Art

A. Coverage path planning

Area decomposition for multi-robot coverage is mature. Divide Areas for Optimal Multi-Robot Coverage (DARP) produces equal-area, connected, robot-specific regions and is the natural reference method [1]. Classical boustrophedon decomposition [2] remains the standard in-region motion primitive for sensor footprints that are rectangular in the body frame.

I adopt a static equal-area partition with boustrophedon in-region coverage rather than a full DARP implementation. This is a deliberate trade of optimality for determinism: a static partition produces an identical, inspectable plan for identical inputs, which is a precondition for the verification regime described in Section V-A. DARP is retained as design rationale and as a future-work path where dynamic reallocation becomes necessary.

B. Multi-robot task allocation

The Consensus-Based Bundle Algorithm (CBBA) is the canonical decentralised auction method for multi-agent assignment [3]. Its guarantees depend on a consensus phase over a communication graph. For a three-aircraft system with a reliable star topology to a single ground station, I implement greedy claim-and-lock with a deterministic tie-break, which is materially simpler to verify and whose worst-case behaviour on three agents is well within acceptable bounds.

C. Aerial detection of people

Small-object detection from altitude is the governing difficulty: a supine adult at survey altitude occupies a few tens of pixels. Slicing-aided hyper inference (SAHI) and related tiling strategies substantially improve small-object recall by running detection on overlapping crops rather than a downsampled full frame [4]. Purpose-built aerial person datasets such as SARD [5] and the broader VisDrone benchmark [6] provide the transfer-learning base; neither contains post-flood Indian terrain, which is why a field data campaign is a funded line in this programme rather than an assumption.

D. Airborne payload delivery

The dominant error term in airborne delivery is release state — velocity and wind — rather than release altitude. Mathisen *et al.* demonstrate autonomous ballistic airdrop from a small fixed-wing UAV and report a mean delivery error of 5.5 m, with the release state computed dynamically against wind and vehicle velocity and re-optimised in flight precisely because that term dominates [7]. This is the decisive argument for a multirotor in this role: a rotorcraft can null its groundspeed to near zero before release and thereby remove the dominant error term outright, which a fixed-wing cannot. The delivery budget in Section IV-F exploits exactly this.

E. Positioning

Real-time kinematic correction from a locally surveyed base is standard practice and is the only route to sub-metre relative positioning without an external network [10]. Since the system may not use internet or cellular services, a team-owned base station is not merely preferable but required.

TABLE I
AIR VEHICLE DESIGN POINT (PER AIRCRAFT)

Parameter	Value
Configuration	Quadrotor
Rotor diameter	18 in
Maximum take-off mass	6.36 kg
Fleet mass (3 aircraft)	19.08 kg
Regulatory fleet cap	25 kg
Battery	6S3P Li-ion 21700, 292 Wh
Cells per aircraft	18
Static thrust-to-weight	2.0
Hover endurance	15.3 min
Design mission duration	7.7 min
Survey altitude	60 m AGL
Ground sample distance	$\leq$2.0 cm/px
Survey groundspeed	8 m/s
Payload	4 kits $\times$ 200 g
Geofence radius	600 m

IV. PROPOSED SYSTEM ARCHITECTURE

A. *Design point*

Table I states the aircraft design point. Every value is produced by a sizing model held in the repository and regenerated in continuous integration; none is transcribed by hand.

The thrust-to-weight ratio of 2.0 and the hover endurance of 15.3 minutes against a 7.7-minute design mission are both reserve policies rather than performance targets: the mission is sized to be completed comfortably within half the available energy, because the failure mode that ends a competition or a deployment is an aircraft that does not return.

Fig. 1 gives the mass statement behind those figures. Two features drive the rest of the design. Structure and battery together are 46 % of maximum take-off mass, which is why airframe mass fraction and pack chemistry are the two sensitivities carried in the risk register rather than, say, avionics choice. The survivor kits are 800 g — 12.6 % of MTOW and fixed by the mission at four kits of exactly 200 g — so payload is an input to the sizing loop, not an output of it. This is the reason the cost floor in Section VII cannot be lowered by component substitution: propulsion and energy are sized by a payload that is not negotiable.

B. *Segment architecture*

The system comprises three segments, shown in Fig. 2.

1) Air segment: Three identical quadrotors, each carrying a Pixhawk-standard autopilot running ArduPilot [9], a dual-band NavIC/GNSS receiver pair for position and moving-baseline heading, a nadir-mounted 12 MP camera, an edge inference module, a four-station kit magazine with positive-lock detents, a mesh radio for telemetry and video, and an independent sub-gigahertz safety link.

2) Ground segment: A single ground control station presenting one unified operator interface. The station is architecturally incapable of originating a retask, waypoint change or drop command; abort and recall are the only permitted downlink commands, and are implemented as an explicit

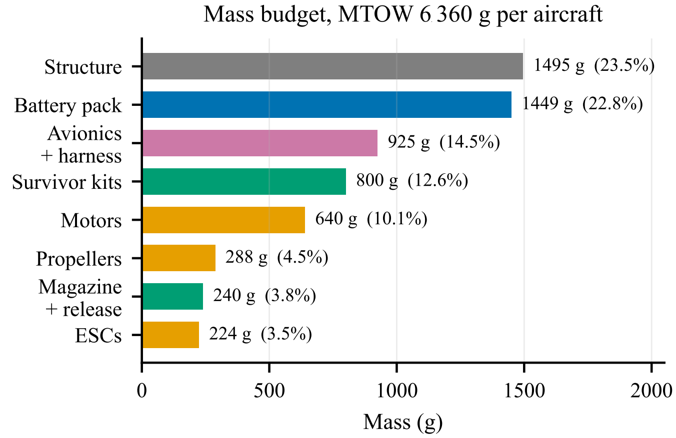

Fig. 1. Mass budget per aircraft. Generated from `docs/sizing/model-output.txt`; the fleet figure of 19 081 g leaves 26 % build-overweight tolerance against the 25 kg regulatory cap.

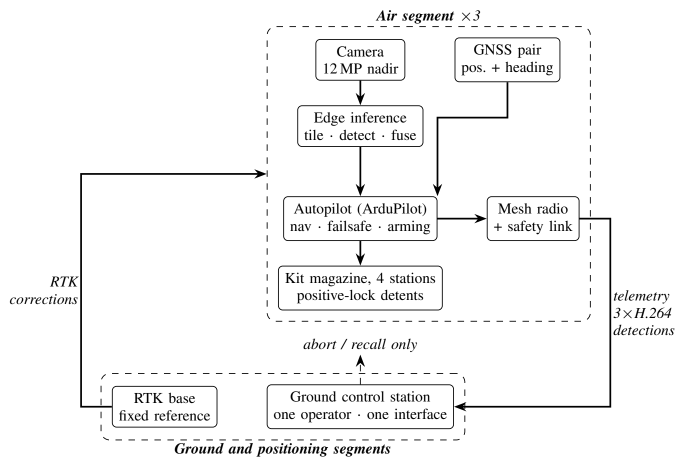

Fig. 2. System architecture. The only permitted downlink commands are abort and recall; the ground station is architecturally incapable of originating a retask, waypoint change or drop command. Detection runs onboard, so the downlink serves the operator rather than the algorithm.

exception. This is a structural safety property rather than a policy: the relevant code paths are absent from the deployed build, not merely guarded.

3) Positioning segment: A team-owned RTK base station, positioned and set to a fixed reference from its first three-dimensional fix within the setup window. Launch is gated on a three-dimensional fix rather than an RTK fix; the first geotag is gated on RTK-fixed, and detections made before convergence are geotagged in float and re-fused once fixed. This decoupling recovers approximately ninety seconds of setup time that a survey-in procedure would otherwise consume.

C. *Geolocation error budget*

Survivor coordinate accuracy is the primary system output, and the error budget is therefore the governing analysis of the whole design. Total error is the root sum of squares of GNSS position error, boresight and lever-arm residual, attitude error, terrain-model error and timing skew.

TABLE II
GEOLOCATION ERROR BY CONFIGURATION

Case	Configuration	RSS error
A	No RTK, single-frame	3.09 m
B	RTK, single-frame	$\sim$1.3 m
C	RTK + multi-frame fusion + calibrated ground plane	**0.91 m**

Table II shows the dominant conclusion: positioning quality, not optics, governs. This is what makes RTK a scoring requirement rather than an optimisation, and it is why the two GNSS receivers survive every cost-reduction pass in Section VII.

Multi-frame fusion is effectively free. At 60 m altitude and 8 m/s groundspeed, forward overlap yields on the order of fourteen independent observations of each target per pass even at a conservative 2 Hz inference rate. Those observations are used both for temporal confirmation, which suppresses false positives, and for geolocation averaging.

Fig. 3 shows both effects quantitatively. Panel (a) makes the positioning argument unarguable: losing the RTK fix moves the error by nearly an order of magnitude, and a 3-D-only fix at 3.67 m RSS consumes most of the 5 m delivery allowance before ballistics or position-hold error are counted. Panel (b) shows why fusion is worth having but not worth over-investing in — it divides the random terms by $\sqrt{n}$ and leaves boresight and target extent untouched, so the curve departs from the ideal $1/\sqrt{n}$ line and flattens by about five frames. Twenty frames is not twice as good as five, and the roughly fourteen frames the geometry supplies for free sit comfortably past the knee.

D. Perception and inference budget

A 12 MP sensor at 60 m yields a ground sample distance of approximately 2 cm/px, at which a supine adult subtends roughly fifty pixels — small, but within reach of tiled inference. Running detection on the full-resolution frame would require 48 tiles per frame and is far beyond the thermal and compute envelope of any module in this mass class. A two-times downsample with 20 % tile overlap requires 12 tiles per frame; at a 2 Hz inference rate this is 24 inferences per second at 640×640, which is within the demonstrated throughput of a 20-TOPS-class edge module.

This is the single most benchmark-sensitive figure in the design. Published throughput for comparable models on comparable hardware spans roughly 24 to 41 frames per second, and the lower end of that range leaves no margin. Hardware benchmarking before design freeze is accordingly a funded, scheduled activity rather than an assumption.

E. Communications

Regulatory rules require the ground station to display a live camera feed from *each* aircraft. Three independent MJPEG streams would demand 4.5–6 Mbps against a link budget of approximately 2.5 Mbps. The system therefore uses H.264 encoding served through a media gateway, carrying three concurrent WebRTC streams in approximately 2.7 Mbps, with per-feed rate reduction under congestion rather than feed switching. Detection runs onboard; the downlink serves the operator, not the algorithm.

An independent sub-gigahertz link in the 865–867 MHz Indian delicensed short-range-device band provides a command and safety path that does not share a failure mode with the primary mesh.

F. Payload delivery

Kits are released at 6 m above ground level, gated on a residual groundspeed below 0.3 m/s. At that release state, ballistic dispersion contributes approximately 0.32 m to total delivery error; combined with geolocation and position-hold error the root-sum-square is approximately 1.5 m with RTK active. Release altitude is chosen low enough that wind drift remains under a metre in realistic conditions, and high enough for safe clearance above a person and above debris.

Positive-lock mechanical detents on each station ensure that a power interruption cannot release a kit. This is treated as a safety-critical item and is excluded from cost reduction by policy.

G. Mission sequence

Fig. 4 shows the autonomous mission sequence. Two properties are worth drawing out. First, the launch stagger lives in the mission file as an explicit delay before each take-off command, not in operator timing — this is what converts a 1.31 m closest approach into the 64.80 m of Fig. 5. Second, every terminal branch converges on return-to-launch: link loss, low battery and operator abort share one recovery path rather than three.

V. METHODOLOGY AND WORK PLAN

The programme is organised into ten phases. Phases P1–P5 are analysis, procurement and integration; P6–P10 are progressively more demanding flight test. The eight-week flight-test window spanned by P6–P10 is the binding schedule constraint on the entire programme, and the work plan is arranged to protect it.

A. Verification strategy

The programme's verification discipline is, I believe, its most transferable contribution, and it exists because of a specific and recurring class of defect.

On four separate occasions during preparatory work, a configuration artefact was found to be correct, unit-tested, reviewed, and *not in the execution path* of the running system. In one instance a validated failsafe parameter set was never loaded by any simulation, so every test ran against stock defaults in which the low-battery action was disabled. In another, a message-routing configuration started cleanly and forwarded zero messages for weeks. Each of these reviewed clean and had passing tests around it.

The countermeasure adopted is procedural rather than technical: no capability is considered demonstrated until the real artefact has been exercised end to end and the resulting values

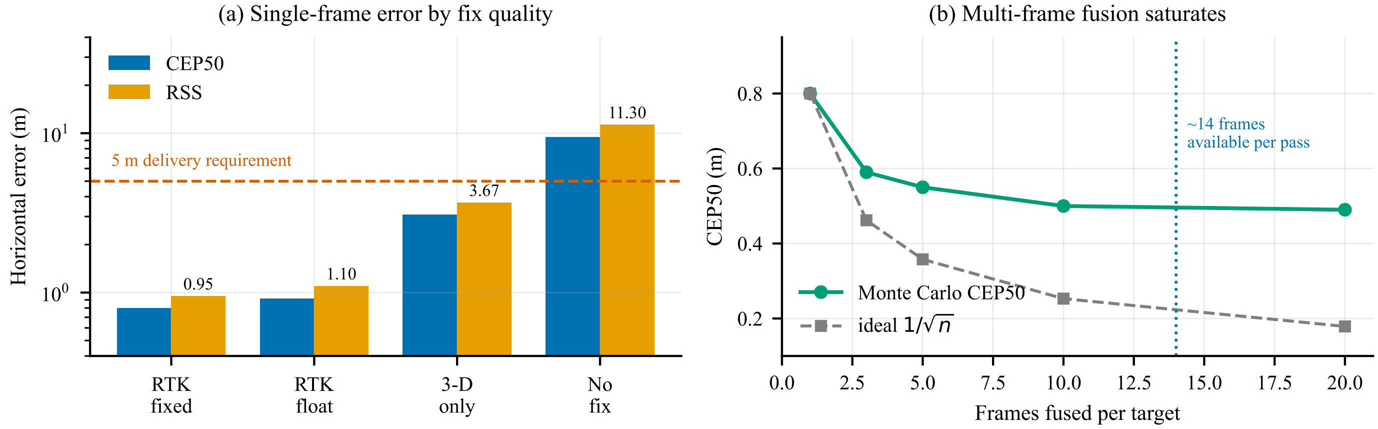

Fig. 3. Geolocation error, from a Monte Carlo run through the same projection code the aircraft flies. (a) Single-frame horizontal error against GNSS fix quality, typical attitude, on a logarithmic scale; the dashed line is the 5 m delivery requirement. (b) Effect of fusing multiple observations of the same target, RTK fixed. Generated by `figures/make_figures.py` from `docs/sizing/geotag-accuracy-output.txt`.

TABLE III
PROGRAMME PHASES

Ph.	Title	Principal exit criterion
P1	Requirements and sizing	Design point frozen; all models regenerate
P2	Ground segment	Single-operator interface demonstrated against three simulated aircraft
P3	Autonomy	Deterministic plan generation; deconfliction verified in simulation
P4	Fault handling	Link-loss and low-battery behaviour verified by fault injection
P5	Procurement and build	Three airframes assembled and weighed; thrust stand complete
P6	First flight	Hover, trim, vibration and thermal characterisation
P7	Perception field trials	Detection recall measured on real imagery with surveyed ground truth
P8	Delivery trials	Delivery CEP measured against a surveyed target mat
P9	Full mission rehearsal	End-to-end mission with all three aircraft, no operator intervention
P10	Setup and contingency	Setup drill timed over repeated runs; margin recovered

read back off the running system. Consequently every claim in this proposal carries an explicit evidence class, and those classes are maintained mechanically rather than editorially:

- **Measured** — observed on a running system, in software or in hardware-in-the-loop simulation.
- **Modelled** — computed by a model whose inputs are themselves assumptions, and which regenerates in continuous integration.
- **Assumed** — neither measured nor modelled; carried explicitly as a risk.

A continuous-integration job enforces the rule that every published number regenerates byte-for-byte from its model, so that changing a model without committing its regenerated output fails the build. Thirteen model outputs are currently under this gate.

VI. Preliminary Results

Substantial work has been completed before this request. I summarise it here with the evidence class of each result stated, because a claim whose basis is declared can be independently checked, whereas a claim whose basis is left implicit cannot be.

A. *Measured in simulation*

- **Low-battery failsafe returns the aircraft to the pad.** Verified across three simultaneous software-in-the-loop autopilots; return-to-launch triggered at 10 809 mAh of a 10 800 mAh planned consumption.
- **Three aircraft on one ground station.** A single telemetry router carried three distinct system identifiers to one ground station port with all three arriving.
- **Inter-aircraft separation at launch.** Staggered take-off delays encoded in the mission file, rather than in operator timing, increased the closest airborne approach during launch from 1.31 m to **64.80 m** (Fig. 5). Both figures are recomputed directly from the committed telemetry.
- **Separation during the sweep.** Minimum airborne separation away from the pad is **29.19 m**, with staggered transit altitudes of 20/25/30 m providing vertical deconfliction independent of the lateral plan.
- **Recovery deconfliction.** Assigning landing slots to the corners of the recovery pad rather than in a row increased slot spacing from 1.22 m to 2.61 m at no cost, raising the measured minimum approach from an overlapping 0.83 m to **2.27 m** — a clearance of 1.22 m between airframes (Fig. 6). Separation while stacked in the air over the pad improves by the same mechanism.
- **Ground station.** All eight required operator displays were verified against three simultaneous autopilots, car-

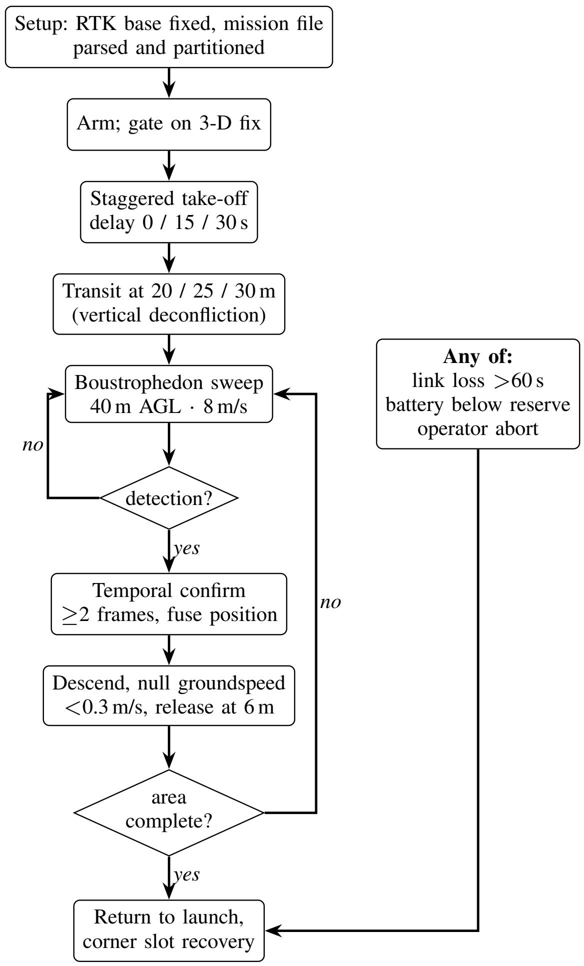

Fig. 4. Autonomous mission sequence. The take-off stagger is encoded in the mission file rather than in operator timing, and all three abnormal conditions converge on a single return-to-launch path.

rying three concurrent H.264 streams within the link budget, with offline map tiles (Fig. 8).

B. Modelled

Endurance, hover power, mass budget, thermal margin, link budget, delivery dispersion and geolocation error are all modelled and regenerate from source. The geolocation Monte Carlo reconciles with the independent analytic budget to within one per cent, and the geotag projection has been verified self-consistent between two independent formulations to 7.8×10^{-10} m.

C. Assumed, or not yet implemented

Detection recall on real post-flood imagery, boresight calibration residual, achieved RTK accuracy in the field, and battery internal resistance under Indian ambient conditions are all assumed. Each requires hardware and is a funded activity in P5–P8.

One ground-segment function is explicitly *not* implemented and is visible as such in Fig. 8: the abort and recall controls currently record operator intent in the mission log but transmit nothing, because no safety radio is yet configured. Recovery today depends on the safety pilot's transmitter. Closing this is a P4 activity and is gated on the sub-gigahertz link discussed in Section IV-E.

D. The honest summary

Nothing in this programme has flown. Every "measured" result above was measured in simulation or in software. That rules out whole classes of logical and integration error and rules out none of the physical ones. The requested funding exists principally to convert the modelled and assumed rows of this section into measured ones.

TABLE IV
AIR VEHICLE BOM BY SUBSYSTEM, OPTION A (PER AIRCRAFT)

Subsystem	**INR**	**Ind.**	**Governing driver**
Avionics	85 400	60 %	Autopilot, RTK pair
Compute, perception	79 500	33 %	Inference throughput
Propulsion	57 396	89 %	Thrust at MTOW
Power	24 500	58 %	292 Wh reserve
Structure	20 550	79 %	Arm root moment
Communications	17 800	42 %	Three-feed budget
Payload system	5 400	61 %	Four stations
Total	**2 90 546**	**58 %**	

VII. Budget and Cost Analysis

A. Basis of estimate

The cost basis is arithmetic over a bill of materials in which each line carries an exact manufacturer part number, an Indian supplier, and a source. Lines are classified by price provenance: *listed* where a dated Indian retail price was obtained, and *quote required* where no public price exists. I regard the distinction as material. Four of the six largest lines in the aircraft — the autopilot, the motor, the cells and the camera — have no public price and require supplier quotation, and the cost basis carries that uncertainty explicitly rather than concealing it in a rounded figure.

B. The bill of materials in detail

Table IV gives the air-vehicle bill of materials at subsystem granularity, with the indigenous fraction each subsystem carries. The full line-level sheet holds 41 entries, each with a manufacturer part number, an Indian supplier and a source link; it is summarised here rather than reproduced.

Two subsystems dominate and they pull in opposite directions. Avionics is the largest line and is also comparatively indigenous, because a Pixhawk-standard autopilot is manufactured domestically. Compute and perception is the second largest and the least indigenous, at 33 %, for the single reason developed in Section VIII: no Indian accelerator exists in this class.

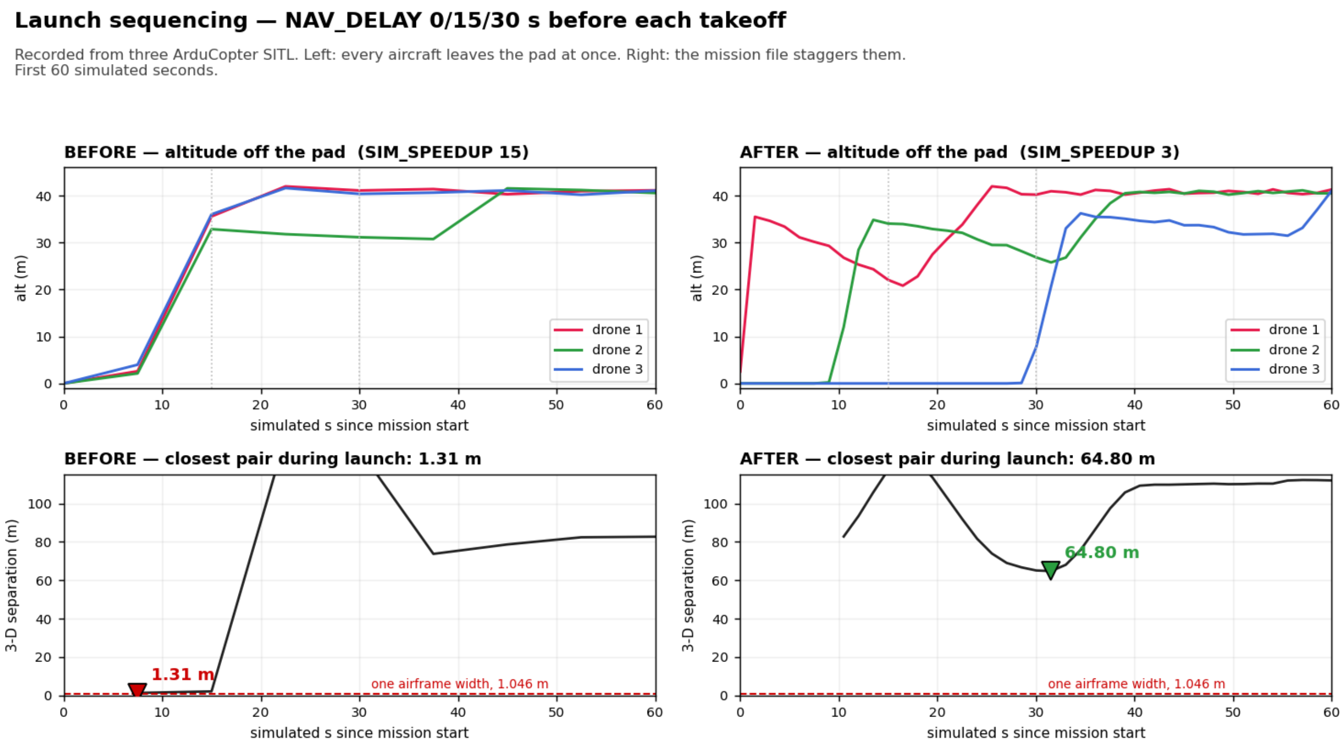

Fig. 5. Launch deconfliction, recorded from three ArduCopter software-in-the-loop autopilots. Left: all three aircraft leave the pad together and the closest pair passes within 1.31 m — inside one airframe width. Right: a delay encoded in the mission file before each take-off command staggers them, and the closest airborne pair is 64.80 m. Both figures are reproducible from the committed telemetry.

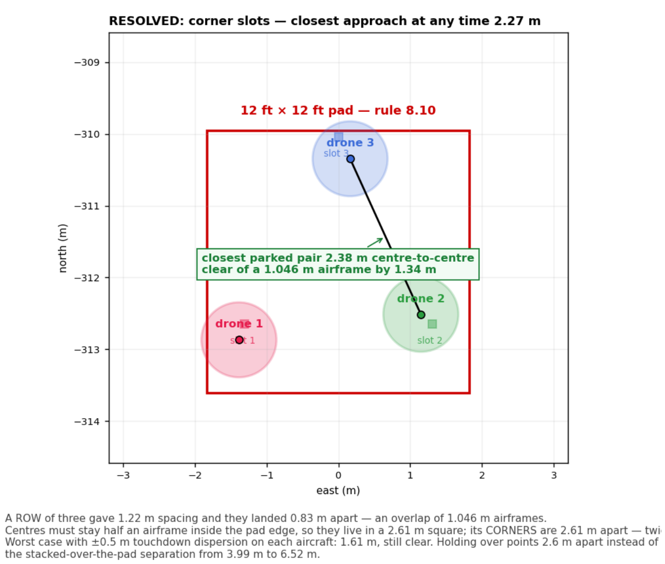

Fig. 6. Recovery deconfliction on the landing pad. Slot centres must remain half an airframe inside the pad edge, so they occupy a 2.61 m square; a row across that square gives 1.22 m of spacing, its corners give the full 2.61 m. The measured closest approach at any time rises from an overlapping 0.83 m to 2.27 m — twice the separation, same pad, no cost.

TABLE V
COSTED CONFIGURATION OPTIONS (PER AIRCRAFT)

Opt.	Description	INR	Capability impact
A	As verified	2 90 546	Baseline
B	**Recommended**	**2 59 001**	None
C	Low cost	2 07 295	Loses heading reference
D	Floor	1 29 632	Loses RTK and inference spec
E	Reduced platform	37 309	Cannot fly the mission

C. Cost reduction study

A structured cost-reduction study was performed against the verified bill of materials, producing four costed configurations of the same aircraft plus one reduced-capability alternative. Table V and Fig. 9 summarise them; Fig. 10 shows where the recommended saving is actually taken.

Option B is the recommended basis. It realises a saving of INR 31 545 per aircraft — INR 94 635 across the fleet — without sacrificing any capability on which the system is assessed. Its principal component is the substitution of the edge inference module for a single-board computer with a 26-TOPS accelerator, which meets the throughput specification at materially lower cost and is conditional on the benchmark described earlier.

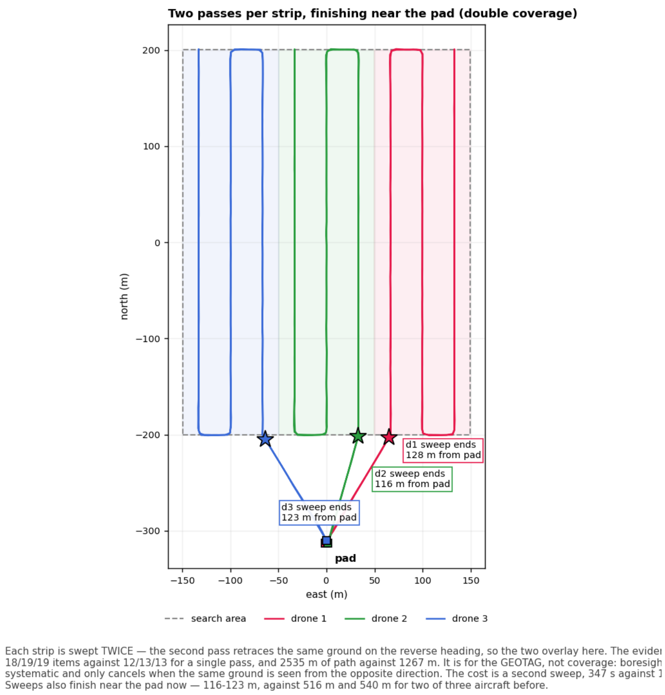

Fig. 7. Coverage decomposition and return geometry. Each aircraft is assigned an equal-area strip and flies a boustrophedon pattern within it. Enumerating the four start-side and direction combinations and keeping the one that ends nearest home brings all three aircraft back within 116–128 m of the pad, against 516 m and 540 m for two of three aircraft under the previous index-keyed rule — at identical path length.

Two findings from the study are worth reporting because they run counter to the usual direction of such exercises. First, the bill of materials was found to be *at or below market* on its largest propulsion line: the carried price was below the lowest listed price for any comparable part, indigenous or imported. Second, the sub-gigahertz safety link was found to be materially under-priced; the carried figure does not purchase a compliant radio for the Indian delicensed band, and the line requires an increase of approximately INR 11 855. Both corrections are incorporated in the figures presented here.

Options D and E are reported for completeness and are not proposed. Option D represents the arithmetic floor for an aircraft that can still fly the mission and is presented as evidence of that floor rather than as a plan; it already sacrifices RTK positioning and the inference specification. Option E is a reduced training platform that cannot carry the payload and is discussed in Section VII-F.

D. Programme cost

I state plainly that this is a development programme rather than a product build. It carries calibration and test equipment, a full spares set including a spare airframe structure, and a field data campaign that a production unit would not. The marginal cost of an additional deployable system is substantially lower than the programme total and is the appropriate figure for any discussion of scaled procurement.

TABLE VI
PROGRAMME COST BREAKDOWN

Element	**INR**
Air vehicles (3)	7.93 L
Payload, ground segment, test equipment, spares and software	11.02 L
Subtotal	18.96 L
Duty and freight, on imported residual	1.34 L
Goods and services tax	3.65 L
Contingency	4.79 L
Total requested	**28.74 L**

TABLE VII
PHASED DISBURSEMENT SCHEDULE

T	**Months / phases**		**INR**	**Release condition**
1	1–2	P1–P4	8.50 L	On award. Design point frozen; models regenerate
2	3–4	P5	9.50 L	Ground segment demonstrated against three simulated aircraft
3	5–6	P6–P8	6.50 L	Three airframes weighed; first hover complete
4	7–8	P9–P10	4.24 L	Perception and delivery trials reported with measured figures
	Total		**28.74 L**	

E. Phased disbursement

The full sum is not required at once. The programme is structured as four tranches tied to the phase exit criteria of Table III, and I would rather be funded against demonstrated progress than hold a sponsor's capital for eight months. Table VII and Fig. 11 give the schedule.

Three properties of this schedule are deliberate.

Tranche 1 is front-loaded and mostly procurement. The largest single schedule risk in the programme is that long-lead components arrive late and compress the flight-test window. Cells, autopilots and GNSS receivers are ordered in month one, before most of the software work is finished, because the lead time is the constraint rather than the readiness of the integration.

Each tranche is released against a stated, checkable exit criterion drawn from Table III — not against a date and not against a progress report. A sponsor can verify every one of them from the artefacts the programme already produces.

Tranche 4 is the smallest and holds the contingency. If the programme underperforms, the unreleased balance is the sponsor's, and the exposure at any point is bounded by the cumulative curve in Fig. 11 rather than by the total.

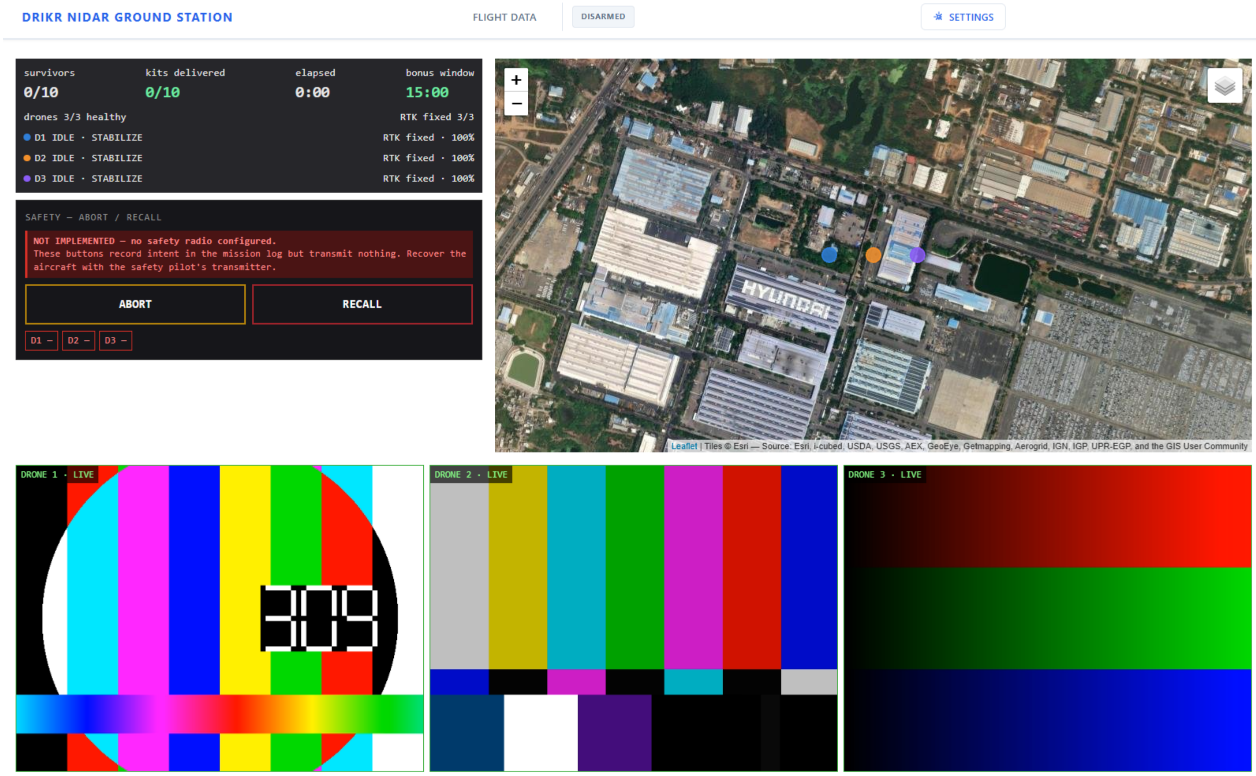

Fig. 8. Ground control station carrying three concurrent H.264 streams within the link budget, with offline map tiles and per-aircraft status. Note the safety panel: abort and recall currently record operator intent in the mission log but transmit nothing, because no safety radio is yet configured. This is reported here rather than cropped out — see Section VI-C.

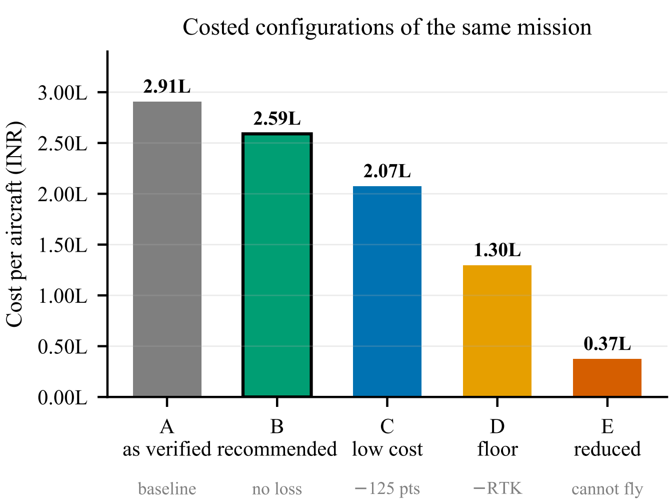

Fig. 9. Costed configurations. A through D are the same aircraft at different prices; E is a different and much smaller airframe that cannot carry the payload, included to show where the cost floor genuinely lies rather than as a proposal.

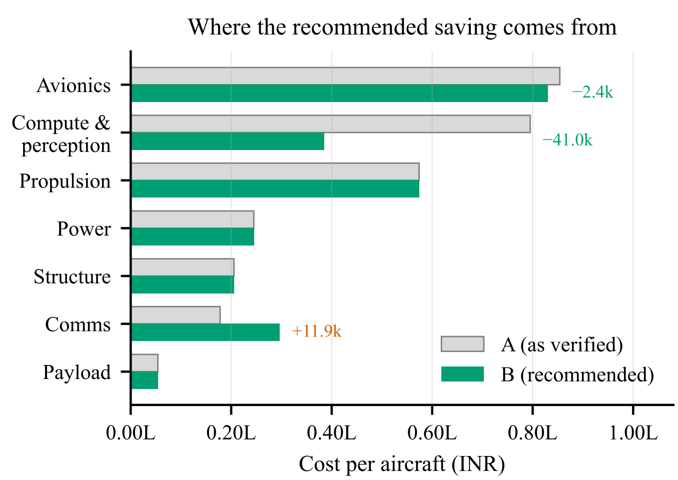

Fig. 10. Subsystem cost, Option A against Option B. Almost the whole saving is one substitution in compute and perception. Communications moves the other way: the sub-gigahertz safety link is corrected *upward* by INR 11 855, because the carried figure does not buy a compliant radio.

F. A note on the training platform

Option E in Table V is a INR 37 309 training airframe. It fails eight system requirements and cannot fly the mission; it is emphatically not a low-cost variant of the proposed system. Its value is different: the programme's binding constraint is an eight-week flight-test window currently budgeted entirely on aircraft that do not yet exist and cannot be risked once they do. Two or three training airframes, at roughly the cost of one inference module, would fly from the first week of that window

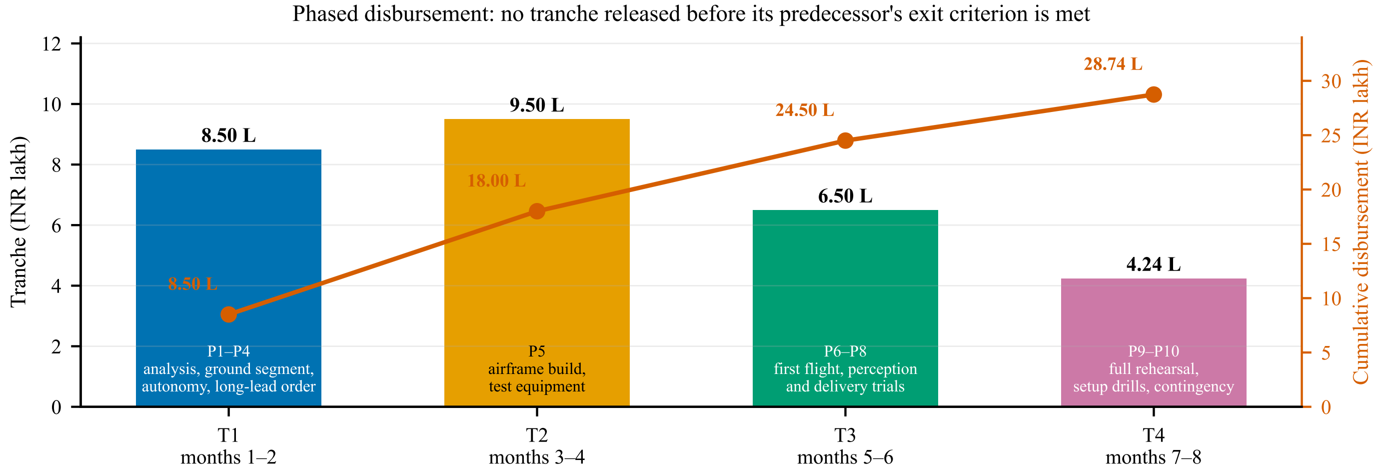

Fig. 11. Phased disbursement against programme phases. Tranche 1 is front-loaded because long-lead components must be ordered in month one for the eight-week flight-test window to survive; tranche 4 is the smallest and carries the contingency.

and absorb ground-station integration, radio-path validation, setup-drill timing, failsafe verification on real hardware, and crash-and-rebuild practice. I identify this as the highest return-on-investment discretionary item in the budget.

VIII. Indigenisation Strategy

Indigenous content is measured, not asserted. Every line in the bill of materials carries a declared Indian value fraction, and the aggregate is computed rather than estimated, which makes the claim auditable.

Value-weighted Indian content across the priced bill of materials is approximately 68 %. I note for precision that weighting the same data by programme spend — in which the air vehicle counts three times — yields approximately 65 %, and I report both rather than the more flattering figure alone. Restricted to the air vehicle alone, the figure is 58 %.

Fig. 12 decomposes that number, and the decomposition is more informative than the aggregate. Indigenous content is not spread evenly: it is high where Indian industry is genuinely strong (propulsion at 89 %, composite structure at 79 %) and low in exactly one place.

The inference accelerator is the one genuinely irreducible import, and I declare it plainly rather than obscuring it. Mitigation is confined to what is honest: an Indian carrier board, Indian enclosure and thermal design, an Indian camera, and a model trained on Indian field data. I regard an explicit gap analysis as more credible than a uniformly confident claim, and I note that the domestic RISC-V development kit carried on the aircraft is a genuine demonstration of Indian silicon in flight, albeit in a monitoring role rather than as an accelerator.

IX. Risk Register

I draw particular attention to the setup-time risk. It is currently supported by a model with a modest margin and no measurement, and it gates the entire mission. Timed bench drills are scheduled as the earliest hardware activity in the programme for precisely this reason.

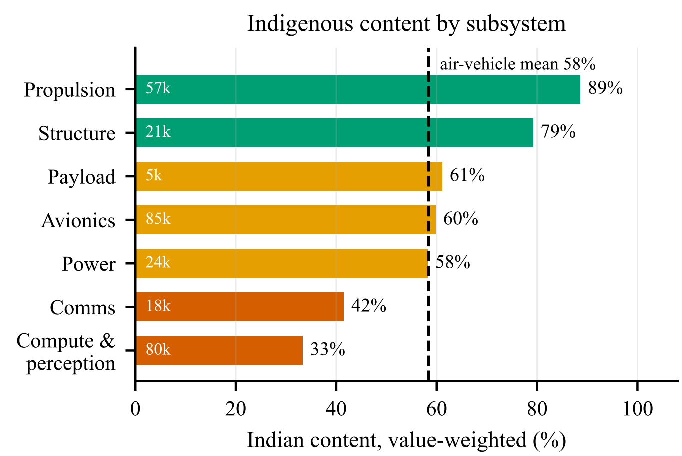

Fig. 12. Indian content by subsystem, value-weighted, with subsystem cost annotated inside each bar. The single subsystem below 40 % is also the second most expensive, which is what makes the aggregate figure worth decomposing rather than quoting alone.

X. Expected Outcomes and Deliverables

1) Three flight-qualified aircraft with a documented, market-verified and predominantly indigenous bill of materials.
2) A single-operator ground control station demonstrated end to end against real aircraft, with no network dependency.
3) A measured detection recall figure on real imagery with surveyed ground truth, replacing the current assumption.
4) A measured delivery circular error probable against a surveyed target, replacing the current model.
5) A measured setup-to-launch time distribution over repeated drills.
6) An open, reproducible evidence base in which every published figure regenerates from its model under continuous integration.

TABLE VIII
INDIGENISATION STATUS BY SUBSYSTEM

Subsystem	Status
Autopilot	Indian, Pixhawk-standard, domestically manufactured
Propulsion	Indian motors, ESCs and propellers with published thrust data
Airframe	Indian composite fabrication
Cells	Indian, BIS-certified 21700 NMC
Camera	Indian embedded-vision supplier
GNSS	Indian receiver with NavIC L1+L5 support
Co-processor	Indian RISC-V development kit
Inference module	**No Indian equivalent exists at this class**

TABLE IX
PRINCIPAL RISKS AND MITIGATIONS

Risk	Sev.	Mitigation
Detection recall below target on real imagery	High	Field data campaign funded in P7; tiling and fusion parameters tunable without hardware change
Inference throughput below 2 Hz on selected hardware	High	Benchmark before design freeze; fallback to reduced tile count on a centre crop
RTK capability unconfirmed on selected receiver	High	Supplier confirmation is a gating procurement action; degraded mode already specified and flagged to the operator
Flight-test window compressed by procurement delay	High	Domestic sourcing halves lead times; training airframes decouple integration from airframe availability
Recovery parachute sizing and supply	Medium	No domestic supplier identified; import lead time and correct mass rating are open procurement actions
Setup time exceeds allowance	Medium	Currently modelled, not measured; timed drills are the highest-priority early test
Airframe loss during test	Medium	Spares set sized for one rebuild; training airframes absorb early-campaign risk

7) A supply-chain report identifying, for each subsystem, the domestic capability that exists, the capability that does not, and what would be required to close the gap.

The seventh deliverable may prove the most durable. The exercise of sourcing a complete airframe domestically, at component granularity and with verified prices, produced a map of Indian UAV supply-chain maturity that I have not seen published elsewhere, including the specific and narrow point at which it currently fails.

XI. CONCLUSION

I propose an autonomous three-aircraft system that addresses the localisation bottleneck in post-flood search and rescue, sourced as far as domestic industry presently permits, and engineered under a verification discipline that distinguishes what has been demonstrated from what has merely been designed.

The system's design point, error budgets and cost basis are established and reproducible. Its coordination and failsafe behaviour have been demonstrated in hardware-in-the-loop simulation across three simultaneous autopilots. What remains is the part that requires funding: to build the aircraft, fly them, and replace the modelled and assumed quantities in Section V-A with measured ones.

I am asking for INR 28.74 L to do that, and I have tried throughout this document to make every number in it checkable.

ACKNOWLEDGMENT

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